Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**
Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, SQL, C/C++
Core Computer Science: Data Structures & Algorithms, OOP, Problem Solving, Multithreading, Exception Handling
Software Development: FastAPI, Flask, REST APIs, API Integration, React.js, JavaScript, HTML, CSS, Git/GitHub
Backend Engineering: API Design, Middleware, Request Validation, Authentication (JWT, OAuth), Concurrent Programming
Databases & Testing: MySQL, SQLite, Redis, SQLAlchemy, PyTest, Unit Testing, Integration Testing, Debugging
Data & Analytics: Pandas, NumPy, Exploratory Data Analysis, Statistical Analysis, Power BI, Data Visualization
AI & GenAI: Machine Learning, NLP, LLMs, RAG, Prompt Engineering, Google Gemini API, Model Evaluation
Development Practices: Test-Driven Development (TDD), Code Review, Performance Optimization, Logging, Error Handling, Technical Documentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and an O(1) LRU cache with lazy/active TTL expiration.
- Designed a four-layer architecture (TCP transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery.
- Validated the system with 46 automated PyTest unit tests (TDD), benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***API Rate Limiter Library* |LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025